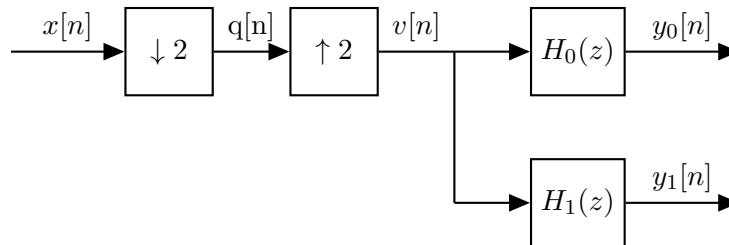


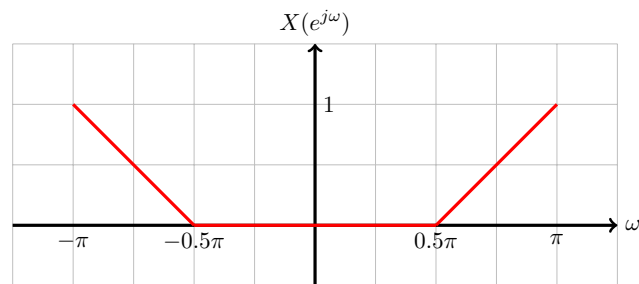
Solutions to Homework #7

Problem 1

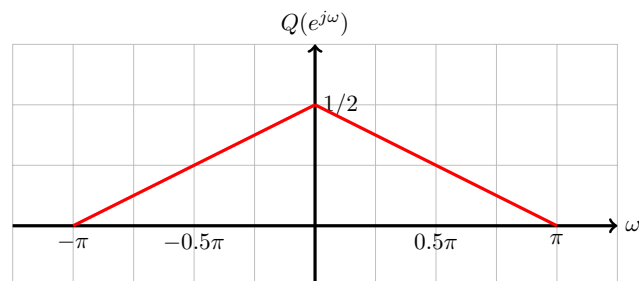
Recall that we consider the system in the following block diagram.



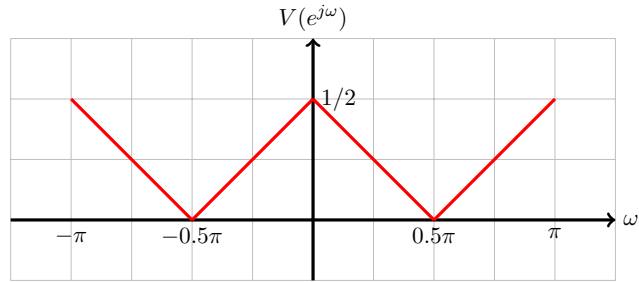
The input signal was given as follows.



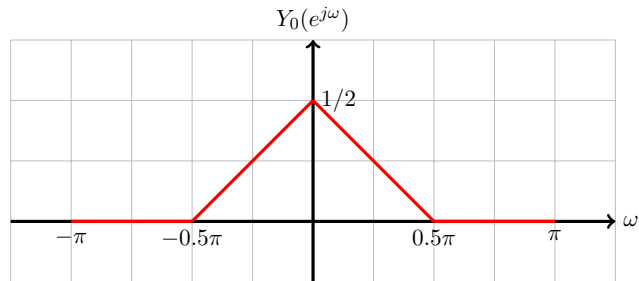
After down-sampling by factor 2, we obtain the DTFT of $q[n]$ as follows.



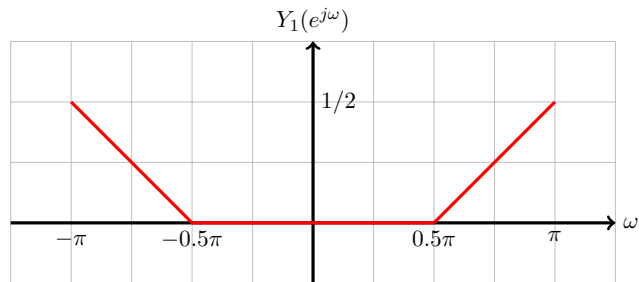
After up-sampling by factor 2, we obtain the DTFT of $v[n]$ as follows.



By applying the ideal LPF $H_0(z)$, we obtain the DTFT of $y_0[n]$ as follows.

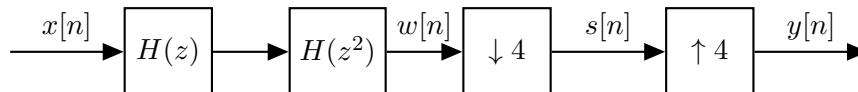


By applying the ideal LPF $H_1(z)$, we obtain the DTFT of $y_1[n]$ as follows.

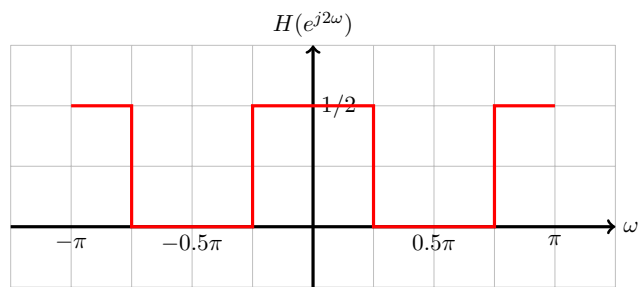
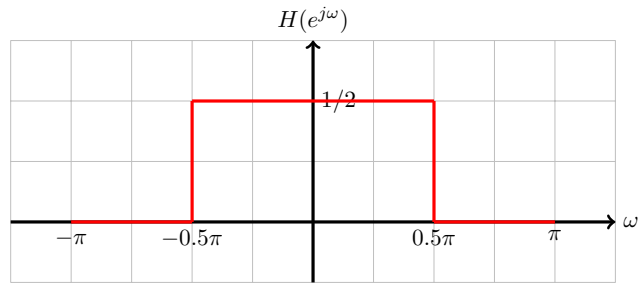


Problem 2

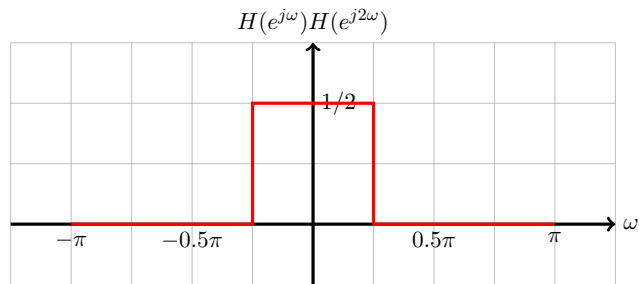
By the noble identity, we obtain an equivalent system given as follows.



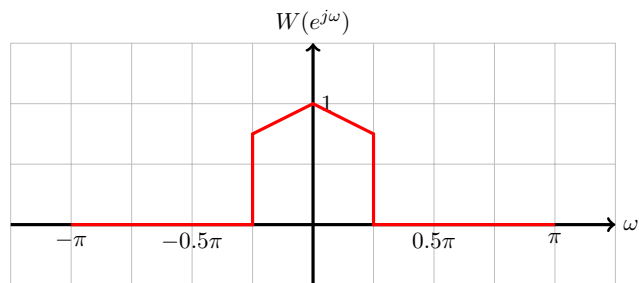
Since



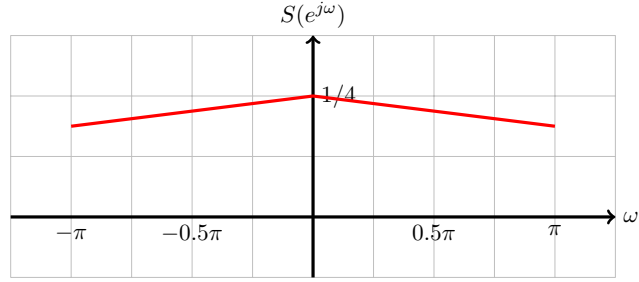
The composite system $H(e^{j\omega})H(e^{j2\omega})$ is given by



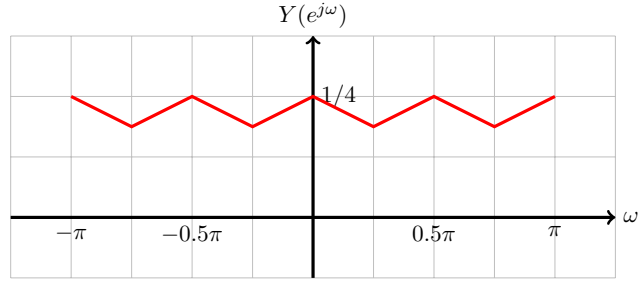
Therefore, the filtered signal $w[n]$ satisfies



The down-sampled signal $s[n]$ satisfies



Finally, the up-sampled signal $y[n]$ satisfies



Problem 3

The first polyphase component is given by

$$h_0[n] = h[2n] = \delta[n] + \left(\frac{1}{2}\right)^2 \delta[n-1] + \left(\frac{1}{2}\right)^4 \delta[n-2] + \left(\frac{1}{2}\right)^6 \delta[n-3] + \left(\frac{1}{2}\right)^8 \delta[n-4].$$

Its z -transform is computed as

$$H_0(z) = 1 + \frac{1}{2^2}z^{-1} + \frac{1}{2^4}z^{-2} + \frac{1}{2^6}z^{-3} + \frac{1}{2^8}z^{-4}.$$

The second polyphase component is given by

$$h_1[n] = h[2n+1] = \left(\frac{1}{2}\right) \delta[n] + \left(\frac{1}{2}\right)^3 \delta[n-1] + \left(\frac{1}{2}\right)^5 \delta[n-2] + \left(\frac{1}{2}\right)^7 \delta[n-3] + \left(\frac{1}{2}\right)^9 \delta[n-4].$$

Its z -transform is computed as

$$H_1(z) = \frac{1}{2} + \frac{1}{2^3}z^{-1} + \frac{1}{2^5}z^{-2} + \frac{1}{2^7}z^{-3} + \frac{1}{2^9}z^{-4}.$$

Problem 4

(a) The first polyphase component is given by

$$h_0[n] = h[2n].$$

Therefore, we have

$$\begin{aligned} H_0(z) &= \frac{1}{2} \left[H(z^{1/2}) + H(-z^{1/2}) \right] \\ &= \frac{1}{2} \left[\frac{a + z^{-1/2}}{1 + az^{-1/2}} + \frac{a - z^{-1/2}}{1 - az^{-1/2}} \right] \\ &= \frac{1}{2} \left[\frac{(a + z^{-1/2})(1 - az^{-1/2}) + (a - z^{-1/2})(1 + az^{-1/2})}{(1 + az^{-1/2})(1 - az^{-1/2})} \right] \\ &= \frac{1}{2} \left[\frac{2a - 2az^{-1}}{1 - a^2z^{-1}} \right] = \frac{a(1 - z^{-1})}{1 - a^2z^{-1}}. \end{aligned}$$

Let $w[n] = h[n + 1]$. Then the second polyphase component is given by

$$h_1[n] = h[2n + 1] = w[2n],$$

which implies

$$\begin{aligned} H_1(z) &= \frac{1}{2} \left[W(z^{1/2}) + W(-z^{1/2}) \right] \\ &= \frac{1}{2} \left[z^{1/2} H(z^{1/2}) - z^{1/2} W(-z^{1/2}) \right] \\ &= \frac{z^{1/2}}{2} \left[\frac{a + z^{-1/2}}{1 + az^{-1/2}} - \frac{a - z^{-1/2}}{1 - az^{-1/2}} \right] \\ &= \frac{z^{1/2}}{2} \left[\frac{(a + z^{-1/2})(1 - az^{-1/2}) - (a - z^{-1/2})(1 + az^{-1/2})}{(1 + az^{-1/2})(1 - az^{-1/2})} \right] \\ &= \frac{z^{1/2}}{2} \left[\frac{2z^{-1/2} - 2a^2z^{-1/2}}{1 - a^2z^{-1}} \right] = \frac{1 - a^2}{1 - a^2z^{-1}}. \end{aligned}$$

(b) The DTFT $H(e^{j\omega})$ is given by

$$H(e^{j\omega}) = H(z)|_{z=e^{j\omega}} = \frac{a + e^{-j\omega}}{1 + ae^{-j\omega}} = \frac{a + \cos \omega - j \sin \omega}{1 + a \cos \omega - ja \sin \omega}.$$

Since we assume that a is real-valued, the magnitude satisfies

$$|H(e^{j\omega})|^2 = \frac{(a + \cos \omega)^2 + \sin^2 \omega}{(1 + a \cos \omega)^2 + a^2 \sin^2 \omega} = \frac{1 + a^2 + 2a \cos \omega}{1 + a^2 + 2a \cos \omega} = 1.$$

Therefore, $H(z)$ is an all-pass filter.

(c) Similar to part (b), the DTFT of $h_0[n]$ is given from $H_0(z)$ as

$$H_0(e^{j\omega}) = \frac{a(1 - \cos \omega + j \sin \omega)}{1 - a^2 \cos \omega + ja^2 \sin \omega}.$$

Its magnitude satisfies

$$|H_0(e^{j\omega})|^2 = \frac{2a^2(1 - \cos \omega)}{1 - 2a^2 \cos \omega + a^4}.$$

The DTFT of $h_1[n]$ satisfies

$$|H_1(e^{j\omega})|^2 = \frac{(1 - a^2)^2}{1 - 2a^2 \cos \omega + a^4}.$$

Therefore, $H_0(z)$ is an all-pass filter if and only if $a = \pm 1$. In this case, $H_1(e^{j\omega}) = 0$ for all ω . On the other hand, $H_1(z)$ is an all-pass filter if and only if $a = 0$. In this case, $H_0(e^{j\omega}) = 0$ for all ω . In general, neither of the two filters is an all-pass filter.